

Curriculum Vitae
Dr. Jatin Majithia

Assistant Professor,
Department of Applied Mathematics and Data Science,
Indian Institute of Information Technology Pune,
Pune: 411048, Maharashtra.



Personal Information

Date of Birth: 08th August 1987

Gender: Male

Nationality: Indian

Mother Tongue: Gujarati

Languages: I am fluent in English, Hindi, Marathi, Gujarati.

Marital status: Married

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Research Interest

Commutative algebra.

Education

Ph.D. in Mathematics from Savitribai Phule Pune University (June 2016- October 2020.)

Thesis Title: Locally nilpotent derivations on affine domains over an algebraically closed field of characteristic zero.

Thesis advisor: Prof. Dr. Dilip Sheth (Principal, SP College Pune).

Co-advisor: Prof. Dr. S. M. Bhatwadekar, INSA Senior Scientist (Retd. TIFR Mumbai).

M.Sc. in Mathematics (July 2010- June 2012)

Department of Mathematics, Savitribai Phule Pune University with 'O' Grade.

B.Sc. in Mathematics (June 2005 - June 2008)

R.Y.K college of Arts and Science, Nashik (affiliated to Savitribai Phule Pune University) with grade "First class with Distinction."

Publications

1. S. M. Bhatwadekar and J. T. Majithia. *ML invariant of the symmetric algebra of a projective module over a one dimensional affine domain*, Journal of Algebra, Volume 498, 336-343, March 2018.
2. S. M. Bhatwadekar and J. T. Majithia. *Class group of the ring of invariants of an exponential map on an affine domain*, Proceedings-Mathematical sciences, Volume 130(1), pages 1-11, Jan 2020.

Ongoing Projects

In collaboration with Prof. S. M. Bhatwadekar, we trying to study kernel of locally nilpotent derivation on $k^{[4]}$.

Teaching / Research Experience

1. Assistant Professor (Sept 2014 - Feb 2022)

I was an Assistant Professor at Department of Mathematics, College of Engineering Pune for 7+ years. **At COEP**, I was:

- (a) Nodal officer for Unnat Bharat Abhiyaan- An initiative by Hon. Prime minister of India. Nodal officer for Unnat Maharashtra Abhiyaan as well. Also member for SES REC.
- (b) Member of Technology and Development cell under Unnat Maharashtra abhiyaan.
- (c) Faculty advisor for Ramanujan Mathematics club from the year 2016 till date.
- (d) Faculty advisor for events in Impressions (Cultural-Fest), Gathering and MindSpark (Tech-Fest). These are yearly events.
- (e) In teaching, average feedback from students till date is ≈ 4.4 out of 5.
- (f) Volunteered for lecture recording for four semesters. Till date recorded around 160 lectures for college.
- (g) In AY 2019-20, from mathematics department, I was a part of the committee which was framed to decide the curriculum structure for the iteration 2019-23 at the institute level.
- (h) Commiittee member at department level for NBA work.

2. Junior research Fellow at IISER Pune: (August 2012-August 2014)

During this period, I did course work of 1 year (Algebra, Topology, Analysis). I also did Mini-Thesis project with Prof. A. Goswami in Financial Mathematics. I also did TA ship for subjects Linear algebra and Calculus.

Internship

Guided one student from NIT Rourkela, who did his summer internship with me for the period of 2 months (14th May to 13th July 2018).

Chapters published

Wrote two chapters (Partial derivatives and Jacobians) for Engineering Mathematics I book for first year students in Savitribai Phule Pune University for FortFlag publication.

Industry Experience

1. Tata consultancy services (October 2008-August 2010).

I was an Assistant system engineer at TCS for 22 months and while leaving the company I was made project leader for the project of the company GE Oil and Gas.

2. ICICI Prudential life insurance (January 2008 to October 2008).

I was financial advisor at ICICI Prudential life insurance after clearing an exam conducted by IRDA.

Scholarships and awards

1. Ph.D Fellowship by National Board of Higher Mathematics, 2012.
2. Cleared CSIR NET for JRF with AIR 114 in August 2012.
3. Received Prof. T. T. Raghunathan scholarship in M.Sc. 2nd year on merit basis.
4. Secured 6th rank in graduation (B.Sc. Mathematics) under Pune university board in 2008.

Courses Taught

Linear algebra, Univariate calculus, Multivariate calculus, Ordinary differential equations, Vector calculus, Laplace transforms, Fourier series.

Other Achievements

1. National cadet corp during the period 2005-08. Cleared NCC 'B' and 'C' certificate examination with 'A' and 'B' grade respectively. Ultimately, got eligible for direct SSB interview for Army.
2. I was "Junior cadet officer (JCO)" in NCC and I was awarded with best turnout of the year 2008.
3. In 2010, I was awarded with green belt certificate in a six sigma project at TCS.

Workshop / conferences (Organiser / delivered lectures)

1. Associate teacher for Algebra (Ring and Module theory) class at AFS II, which was held at Bhaskaracharya Pratisthana, Pune from 16th May 2016 to 12th June 2016. (took tutorial sessions for all four weeks)
2. Associate teacher for Algebra (Ring and Module theory) class at IST, which was held at Bhaskaracharya Pratisthana, Pune from 28th Nov 2016 to 10th Dec 2016.(took tutorial sessions for all four weeks)

3. **Organizer** for the conference on “National conference on Pure and Applied Mathematics”, which was organised by Department of Mathematics, COEP from 17th to 19th May 2017.
4. Associate teacher for Algebra (Ring and Module theory) class at AFS II, which was held at Bhaskaracharya Pratisthana, Pune from 8th May 2017 to 3rd June 2017.(took tutorial sessions for all four weeks)
5. Associate Teacher for Algebra (Group Theory) class at AFS I, which was held at Bhaskaracharya Pratisthana, Pune from 4th-30th Dec 2017.(took tutorial sessions for all four weeks)
6. Attended and Presented a paper at “International Conference in Algebra and Analysis” which was held at SPPU from 19th Dec-22th Dec 2017.
7. Delivered a lecture for the final round of the competition “Genius Junior”. This competition was organised by MindSpark on 30th Sept 2018 after selecting top teams of class 8 students from schools all over Maharashtra.
8. Associate teacher for Algebra (Group Theory) class at AFS I, which was held at Bhaskaracharya Pratisthana, Pune from 4th - 30th Dec 2018.(took tutorial sessions for all four weeks)
9. Associate teacher for Algebra (Ring and Module theory) class at AFS II, which was held at Bhaskaracharya Pratisthana, Pune from 27th May 2019 to 17th June 2019.(took tutorial sessions for all four weeks)
10. **Attended and Presented** a paper at a National Conference which was held at MIT Alandi on 7th June 2019.
11. **Organizer** for the one day workshop on “Semigraph”, which was organised by Department of Mathematics, COEP on 11th January 2020.
12. **Organizer** for three days workshop on “L^AT_EX for Professionals”, which was organised by Ramanujan Mathematics Club and Department of Electrical engineering, COEP from 19th to 21st February 2020. Also delivered a series of talk on ”Math made impressive” in this workshop.
13. **Organised** an event on 5th June 2021 on the occassion of World Environment Day. It was a national event in which three speakers delivered the talk.

Workshops / Conferences / FDPs (Attended)

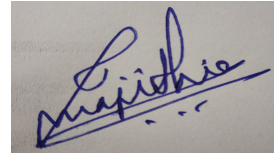
1. “Pedagogy workshop”, organized by Microsoft and College of Engineering, Pune at College of Engineering, Pune on 20th march 2015.
2. “Micro Nano Technology in Service of Rural India”, organized by BARC Mumbai and College of Engineering Pandharpur at Pandharpur on 12th -13th April 2015.
3. “Teaching Applied Mathematics to Post Graduate Students in Engineering”, Which was held at IIT Mumbai from 13th to 15th May 2015.
4. “Mathematica as a Teaching Aid”, organized by IIT Roorkee between 9th -12th March 2016 under TEQIP.

5. “Educational Technology for Engineering Teachers”, organized by IIT Bombay between 7th Jan 2016-8th March, it was an online course.
6. “Induction training program phase I for faculty”, organized by E n TC Department, COEP and NITTTR Bhopal from 27th June-8th July 2016.
7. One day Western Region Workshop organized by CTARA Cell of IIT Bombay on 16th November 2016 for the work to be done under Unnat Bharat Abhiyaan and Unnat Maharashtra Abhiyaan.
8. Two days workshop on “Teaching Strategies in Mathematics” at BITS Pilani Goa on 28th -29th April 2017.
9. Attended CAAG 2017 at IISER Pune from 5th-8th Dec 2017.
10. Attended one week online FDP on Machine learning and Deep learning application in Engineering and Science from 16th May to 20th May 2020.
11. Attended online webinar on CO-PO Attainment computation and Outcome analysis organised by Inpods on 30th May 2020.
12. Attended one week online course on Digital Teaching Techniques organised by ICT Academy from 27th July to 01st Aug 2020.
13. Attended one day online workshop on “Social Entrepreneurship, Swachhta and Rural Engagement’ for Higher Educational Institutions in association with Mahatma Gandhi National Council of Rural Education on 18th Sept 2020.
14. Attended one day online webinar sponsored by NBA on “Outcome based education and Accrediation”, organised by NBA Delhi, DTE Maharashtra and College of Engineering Pune on 27th Nov 2020.
15. Attended TEQIP-III sponsored one week online Short Term course On ”Mathematics Without Boundaries”, Organized by Department of Mathematics, National Institute of Technology, Srinagar during 23-27 November, 2020.
16. Successfully completed four days online workshop on “Train the trainers on Examination reforms”, organised by BVB College of Engineering and Technology, Hubballi in collaboration with NPIU from 4th to 7th December 2020.
17. Attended one day online international conference on “History and Development of Mathematics (ICHDM 2020)”, to commemorate 133rd Birth anniversary of Srinivasa Ramanujan, on 22nd December 2020.
18. Attended Two weeks “Refresher course in Mathematics” organized by department of Mathematics, Ramanujan college from 31st Aug to 14th Sept 2021.
19. Successfully completed NPTEL online certification on “Essential Mathematics for Machine Learning”. The course was run on Swayam Platform by faculty from IIT Roorkee from July 2021 to Oct 2021.

20. Attended 5 days offline FDP on “Data Analytics with Python”, organised by Department of Computer Engineering and IT, College of Engineering Pune from 20th to 24th December 2021.

I hereby declare that the above mentioned facts are true to the best of my knowledge and belief.

Place: Pune
Date: 15/02/2022



Dr. Jatin Majithia